Subject Code: KCE078

PAPER ID-310037

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM VII) THEORY EXAMINATION 2024-25
IRRIGATION AND WATER RESOURCE ENGINEERING

TIME: 3 HRS **M.MARKS: 100**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

Q no.	Question	CO	Level
a.	Define precipitation and what are the major forms of precipitation?	1	K1
b.	Define recording type of rain gauge.	1	K1
c.	State advantages of lined canals.	2	K1
d.	State the necessity of irrigation.	2	K1
e.	Define sediment.	3	K1
f.	Define cross drainage works.	3	K1
g.	Write the cause of meandering.	4	K1
h.	Define a spur.	4	K1
i.	Define the terms aquiclude and aquitard.	5	K1
j.	Define Darcy's law and list out the validity of Darcy's law.	5	K1

SECTION B

2. Attempt any *three* of the following: **10 x 3 = 20**

Q no.	Question	CO	Level
a.	Explain the stages of the hydrologic cycle and their significance in water resource management.	1	K2
b.	Explain the components of a hydrograph and their hydrological significance.	2	K2
c.	Design an irrigation channel to carry 35 cumecs of discharge, with B/D i.e., base width to depth ratio as 2.25. The critical velocity ratio is 1.0 and Kutter's rugosity coefficient is 0.022 and use Kennedy's method.	3	K2
d.	Discuss the objectives of river training.	4	K2
e.	Describe Confined and Unconfined aquifer with suitable diagram.	5	K2

SECTION C

3. Attempt any *one* part of the following: **10 x 1 = 10**

Q no.	Question	CO	Level
a.	Explain any four factors affecting runoff.	1	K2
b.	Describe Theissen polygon method of calculating average rainfall.	1	K2

4. Attempt any *one* part of the following: **10 x 1 = 10**

Q no.	Question	CO	Level
a.	Explain the concept of a Direct Runoff Hydrograph and how it differs from a total hydrograph.	2	K2
b.	How do you estimate consumptive use of crop? Explain in detail.	2	K2

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5. Attempt any *one* part of the following: 10 x 1 = 10

Q no.	Question	CO	Level
a.	Design an irrigation channel based on Kennedy's theory with the following details Discharge : 60 cumec Bed Slope : 1 in 6000 Critical Velocity Ratio m : 1.05 Rugosity coefficient : 0.02	3	K3
b.	Design an irrigation channel using Lacey's theory for the following data Discharge : 40 m3 /sec Silt factor : 1.02 Side Slope : ½ : 1	3	K3

6. Attempt any *one* part of the following: 10 x 1 = 10

Q no.	Question	CO	Level
a.	How are river training works classified? Explain about any two types of river training works in a detailed manner.	4	K2
b.	What are the basic principles of regulation of a canal system? Explain the various method of regulation of canal system.	4	K2

7. Attempt any *one* part of the following: 10 x 1 = 10

Q no.	Question	CO	Level
a.	Derive the expression for the steady state discharge of well fully penetrating into a unconfined aquifer.	5	K3
b.	Illustrate the terms (i) cone of depression (ii) specific yield (iii) flowing well (iv) Darcy's velocity.	5	K3

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